

Beyond Scaling: Bayesian Temporal Inductive Bias for Time-Series Foundation Model Adaptation

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Abstract—Time series foundation models (TSFMs) have shown promising forecasting capabilities by pretraining Transformer architectures on large-scale time-series corpora. Nevertheless, scaling model capacity does not consistently improve forecasting accuracy. Meanwhile, time series are typically non-stationary and exhibit temporal dependencies that vary across inputs and regimes. To investigate how TSFMs model such dependencies, we develop a Bayesian interpretation of the attention distribution, factorizing it into a content-based prior induced by canonical self-attention and a temporal-evidence likelihood induced by positional encoding (PE). This formulation characterizes the essential role of PE and reveals that static positional biases can be suboptimal for non-stationary temporal patterns. We therefore propose an adaptive PE method based on a mixture of temporal experts, enabling pretrained TSFMs to adapt to target-specific local, periodic, and long-range dependencies. Extensive experiments demonstrate improvements in average forecasting performance while maintaining computational efficiency.

Index Terms—time series foundation models, positional encoding, Bayesian attention, mixture of experts, non-stationarity

I. INTRODUCTION

Time series forecasting (TSF) is a fundamental problem in data mining that aims to predict future trajectories over multiple horizons from historical observations [1]. Owing to its critical role in a wide range of real-world applications, such as finance [2] and climate and environmental monitoring [3], TSF has remained a long-standing and active research topic. Inspired by the remarkable success of foundation models in Natural Language Processing (NLP), such as GPT-3 [4] and T5 [5], and Computer Vision (CV), such as ViT [6] and CLIP [7], Time Series Foundation Models (TSFMs) [8] have recently attracted significant attention in the TSF community.

A central principle underlying modern foundation models is the scaling paradigm [9]: increasing model capacity and the scale of pretraining data can systematically improve a model’s ability to learn generalizable representations and transfer them to previously unseen tasks or data distributions. Recent studies [10], [11] have confirmed the scaling paradigm in the time-series forecasting domain. Motivated by this observation, TSFMs are typically designed as large-scale models pretrained on diverse time-series corpora, with the goal of learning broadly transferable temporal representations that support forecasting across heterogeneous datasets and domains with little or no task-specific training.

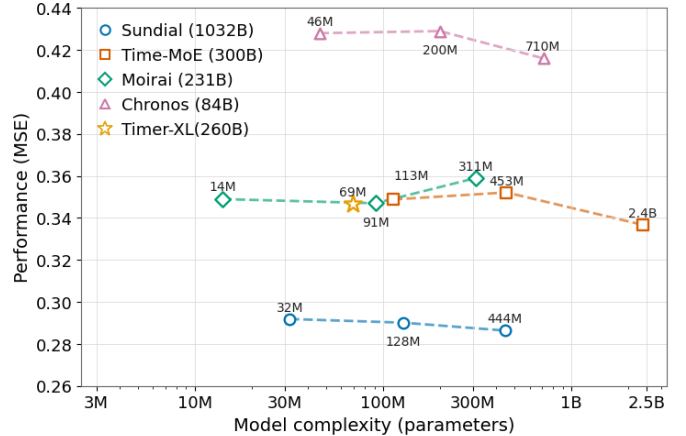


Fig. 1. Zero-shot forecasting performance of different TSFMs pretrained on datasets of different scales. The number after each TSFM in the legend denotes the number of Billions of time points used for pretraining. The plot shows that zero-shot forecasting performance improves only marginally or non-monotonically as model size increases within each TSFM family, whereas larger pretraining time-point scales are associated with more substantial performance gains across different TSFM architectures.

Unlike the pronounced performance gains associated with model scaling in NLP [9], Fig. 1 shows that increasing the complexity of TSFMs does not consistently yield commensurate improvements in forecasting accuracy. One plausible explanation is that the backbone architectures of most TSFMs largely inherit the Transformer architecture developed for NLP, despite fundamental differences between textual and time-series data: text corpora often exhibit relatively stable aggregate statistical laws despite substantial contextual and temporal heterogeneity [12], whereas time series are typically non-stationary. As a result, simply scaling model capacity does not necessarily address this distribution shift challenge. In contrast, target-specific fine-tuning or adaptation provides a direct mechanism for mitigating such distribution mismatch by exposing the pretrained model to observations from the target time series, potentially yielding better performance gains than simply increasing model capacity. Yet, comparatively little work has systematically characterized how Transformer architectures behave under these time-series-specific statistical properties, leaving it unclear how to adapt TSFM architectures.

More fundamentally, the canonical self-attention — the core computational mechanism underlying most TSFMs — is permutation-equivariant and therefore cannot encode temporal order without additional positional information [13], [14]. In other words, temporal dependencies are learned through the interaction between positional encoding (PE) and content-based self-attention in TSFMs. The plateauing performance of TSFMs shown in Fig. 1 therefore raises a fundamental question: how does PE interact with self-attention to model temporal dependencies when those dependencies vary across input windows? Answering this question is essential for understanding how TSFMs perform forecasting under non-stationarity and ultimately for adapting TSFM architectures.

In this paper, we investigate this question from a Bayesian perspective. Specifically, we introduce an auxiliary variable to provide a Bayesian interpretation of the interaction between content-based attention and temporal positional weighting. Under this formulation, the attention distribution in a TSFM can be interpreted as a posterior that factorizes into two components: (i) a content-based prior induced by canonical self-attention and (ii) a temporal-evidence likelihood induced by PE, which characterizes the temporal dependency structure between queries and keys. This formulation provides a unified probabilistic interpretation of commonly used PE methods, including RoPE [15], ALiBi [16], KERPLE [17], and relative positional encoding (RPE) [18], thereby highlighting the essential role of PE in TSFMs.

Moreover, we argue that existing PE methods are suboptimal for TSFMs. Time series are inherently non-stationary, such that their temporal dependency structures can vary substantially across time, samples, and regimes. However, conventional PE methods learn a fixed position-to-bias mapping during training and apply the same positional structure to all input sequences at inference time. Consequently, static positional inductive biases may fail to accurately characterize the evolving temporal patterns of non-stationary time series. This limitation becomes particularly pronounced in TSFMs pretrained on large-scale, diverse, and heterogeneous time-series corpora, where temporal dynamics can vary substantially across datasets, domains, and instances.

Inspired by the recent success of mixture-of-experts (MoE) models in NLP [19] and TSF [20], [21], we propose an adaptive PE method for fine-tuning pretrained TSFMs. The method first constructs a temporal-evidence profile for each attention head by aggregating the original attention scores according to time lag and augmenting the resulting profile with its Fourier spectrum. An MoE router then assigns input- and head-specific weights to a set of predefined temporal kernels, whose weighted combination produces an adaptive bias for the attention logits. This design enables different inputs and attention heads to emphasize distinct local, periodic, and long-range dependency patterns while retaining the standard multi-head self-attention computation. Consequently, the proposed method can adapt the positional inductive bias of a pretrained TSFM to the temporal characteristics of a target dataset.

Overall, our main contributions are summarized as follows:

- We develop a Bayesian interpretation of self-attention for time-series forecasting, in which the attention distribution is factorized into a content-based prior induced by canonical self-attention and a temporal-evidence likelihood induced by PE. This formulation provides a unified probabilistic interpretation of widely used PE methods and theoretically characterizes the essential role of PE in enabling TSFMs to capture temporal dependencies.
- We reveal a fundamental limitation of static PE methods for time-series forecasting. Because time series are typically non-stationary, their temporal dependency structures may vary across time, samples, attention heads, datasets, and regimes. Consequently, a fixed positional bias shared across all inputs can be suboptimal for adapting TSFMs to heterogeneous and evolving temporal patterns.
- We propose an adaptive PE method based on a mixture of temporal experts. An MoE router dynamically combines predefined temporal kernels using the temporal evidence extracted for each input and attention head, allowing the positional bias to adapt to local, periodic, and long-range dependencies. Experiments on five benchmark datasets demonstrate average improvements in forecasting performance while maintaining computational efficiency.

II. RELATED WORK

A. Time-Series Foundation Model Adaptation

TSFMs learn transferable temporal representations through pretraining on diverse time-series corpora [8]. Representative models include Moirai [22], Chronos [23], Timer and Timer-XL [24], [25], Time-MoE [20], and Sundial [26]. Existing studies adapt TSFMs through online forecast correction, in-context examples, or parameter-efficient fine-tuning [27]–[29]. These approaches do not explicitly address adaptation through the positional mechanisms used to represent non-stationary temporal dependencies. Our work focuses on this aspect, using a Bayesian interpretation to motivate input- and head-dependent positional biases for target-specific TSFM adaptation.

B. Positional Encoding Methods for Transformers

Common positional encoding (PE) methods include absolute positional encoding (APE) [30], relative positional encoding (RPE) [14], ALiBi [16], KERPLE [17], and RoPE [15]. These methods incorporate positional information into input representations or attention scores. However, their positional mechanisms are generally fixed after pretraining and are not explicitly designed to adapt to time-series dependency patterns that vary across samples and temporal regimes [13], [31]. Our method instead uses input- and head-specific temporal evidence to dynamically combine kernels emphasizing local, periodic, and long-range dependencies.

C. Mixture-of-Experts (MoE)

MoE models use a router to assign input-dependent weights to multiple experts, enabling specialization in different patterns. In time-series forecasting, MoE combines lightweight

linear forecasting experts [32]; Pathformer adaptively selects pathways at different temporal scales [33]; and DUET routes heterogeneous temporal distributions to specialized pattern extractors [34]. DTAF employs MoE filters to disentangle and suppress changing temporal components for non-stationary forecasting [35]. Time-MoE [20] and Moirai-MoE [21] incorporate sparse expert routing into TSFMs to increase model capacity while activating only a subset of parameters. Whereas these methods primarily use MoE for feature extraction, forecasting, or model scaling, we use it to combine temporal kernels and adapt the positional inductive bias of pretrained TSFMs.

III. A BAYESIAN EXPLANATION FOR SELF-ATTENTION

In this section, we develop a Bayesian interpretation of self-attention to clarify the role of positional encoding in modeling temporal dependencies and motivate its adaptation to non-stationary time series.

Based on the kernel interpretation [36], [37] of multi-head scaled dot-product self-attention [30], the attention assigned by the i th query $\mathbf{q}_i$ to the j th key $\mathbf{k}_j$ is defined as

$$\alpha_{ij} = \frac{\exp(\mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d_k} + b_{ij})}{\sum_{l \in \mathcal{C}(i)} \exp(\mathbf{q}_i^\top \mathbf{k}_l / \sqrt{d_k} + b_{il})}, \quad (1)$$

where $\mathbf{q}_i$, $\mathbf{k}_i$, and $\mathbf{v}_i \in \mathbb{R}^{B \times H \times d_k}$ are the query, key, and value at position $i \in \{1, \dots, T\}$, respectively, and denote different representations of the self-attention input. In particular, b_{ij} denotes the inductive bias introduced by PE. $\mathcal{C}(i)$ denotes the set of keys accessible to query i ; under causal attention, $\mathcal{C}(i) = \{1, \dots, i\}$; otherwise $\mathcal{C}(i) = \{1, \dots, T\}$.

Equation 1 indicates that $\mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d_k}$ measures the content-based similarity between the query and key, whereas b_{ij} represents the positional bias induced by positional encoding (PE). The positional bias modulates the attention logit according to the positions of the query and key, thereby injecting information about temporal order, relative lag, or other predefined temporal structures into the attention distribution. Without PE, unmasked self-attention is permutation-equivariant [31], [36], [38]. For a fixed query, content-only attention cannot distinguish identical keys within its accessible key set. In particular, if $j, j' \in \mathcal{C}(i)$ and $\mathbf{k}_{j'} = \mathbf{k}_j$ for $j' \neq j$, then $\alpha_{ij'} = \alpha_{ij}$ when $b_{ij} = 0$, because their content-based attention logits are identical. PE resolves this ambiguity by assigning position-dependent biases b_{ij} and $b_{ij'}$, allowing self-attention to distinguish identical or similar observations occurring at different temporal positions.

Let $s_{ij} = \mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d_k}$ denote the content-based attention score. We distinguish the attention weights α_{ij} from the content-based prior π_{ij} , defined by

$$\pi_{ij} = \frac{\exp(\mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d_k})}{\sum_{l \in \mathcal{C}(i)} \exp(\mathbf{q}_i^\top \mathbf{k}_l / \sqrt{d_k})}.$$

The positional bias b_{ij} induces a normalized temporal relevance distribution over accessible key positions:

$$\rho_i(j) = \frac{\exp(b_{ij})}{\sum_{l \in \mathcal{C}(i)} \exp(b_{il})}.$$

Because $\log \rho_i(j)$ differs from b_{ij} only by a query-specific constant, replacing b_{ij} with $\log \rho_i(j)$ leaves the attention weights unchanged.

To obtain a Bayesian interpretation, introduce a binary auxiliary variable E_i such that

$$\Pr(E_i = 1 \mid J_i = j, \mathbf{X}) = \rho_i(j),$$

where J_i denotes the selected key index and $\mathbf{X}$ denotes the input. The content-based prior is $\Pr(J_i = j \mid \mathbf{X}) = \pi_{ij}$.

Conditioning on the same event $E_i = 1$ for all candidate keys, Bayes' rule gives

$$\Pr(J_i = j \mid E_i = 1, \mathbf{X}) = \frac{\pi_{ij} \rho_i(j)}{\sum_{l \in \mathcal{C}(i)} \pi_{il} \rho_i(l)} = \alpha_{ij}. \quad (2)$$

Consequently,

$$\log \alpha_{ij} = s_{ij} + \log \rho_i(j) - \log A_i, \quad A_i = \sum_{l \in \mathcal{C}(i)} \exp(s_{il}) \rho_i(l).$$

Thus, setting $b_{ij} = \log \rho_i(j)$ recovers attention with an additive temporal bias. The auxiliary variable provides a probabilistic interpretation of this weighting mechanism; it does not represent an independently observed temporal label.

In particular, RoPE [15] can also be rewritten as an additive correction to the content-based score:

$$\frac{\mathbf{q}_i^\top \mathbf{R}_{j-i} \mathbf{k}_j}{\sqrt{d_k}} = s_{ij} + \frac{\mathbf{q}_i^\top (\mathbf{R}_{j-i} - \mathbf{I}) \mathbf{k}_j}{\sqrt{d_k}},$$

where $\mathbf{R}_{j-i} = \mathbf{R}_i^\top \mathbf{R}_j$ denotes the relative rotation matrix. The Bayesian perspective also relates positional biases to temporal weighting: ALiBi [16] induces exponential decay with lag, whereas the logarithmic variant of KERPLE [17] and the power-law bias in Powerformer [39] induce power-law weighting. For RoPE, the additive correction depends on both content and relative position, rather than lag alone.

The Bayesian interpretation above highlights a limitation of static lag-based positional biases. For methods such as ALiBi and KERPLE, the bias takes the form $b_{ij} = f_\theta(i - j)$, with the same learned or predefined function applied to every input window after pretraining. The corresponding temporal-evidence distribution therefore imposes a fixed lag-dependent weighting. Although θ may differ across attention heads, it does not adapt to the temporal characteristics of each input. However, time series are typically non-stationary: the relevance of the same lag can change substantially across input windows. For example, a recent observation may dominate during a rapidly changing trend, periodic lags may be more informative under strong seasonality, and long-range observations may become important in slowly evolving regimes. The content-based self-attention prior may not fully compensate for such variation, as it measures query-key similarity without an explicit adaptive lag-dependent bias. Consequently, a mismatched static temporal weighting can bias the resulting attention posterior toward temporal patterns that do not match the current input. An adaptive PE should therefore condition the positional bias on both the input and the attention head, i.e., $b_{ij}^{(n,h)} = f(i - j; \mathbf{X}^{(n)}, h)$, where $\mathbf{X}^{(n)}$ denotes the n th input

time-series window and h denotes the attention head index, thereby allowing self-attention to emphasize different local, periodic, or long-range dependencies for different time-series windows. This observation motivates our Bayesian temporal inductive bias, which adaptively adjusts the temporal relevance distribution to facilitate target-specific TSFM adaptation.

IV. METHODOLOGY

Given a TSFM pretrained with static positional encoding, we adapt it to a target dataset in two steps. First, we replace the pretrained static PE mechanism in each self-attention layer with the proposed Bayesian temporal inductive bias, while retaining the remaining pretrained parameters as initialization. Second, we fine-tune the modified TSFM on the target dataset by jointly optimizing the MoE router and the TSFM parameters under the original forecasting objective. This strategy builds on the transferable content representations learned during pretraining while adapting how the model represents temporal dependencies.

The Bayesian temporal inductive bias consists of two components: (A) a probabilistic mixture of time-aware kernels, where each kernel represents a distinct temporal inductive bias; and (B) an MoE router that produces input- and head-specific weights for combining these kernels. The logarithm of the resulting temporal relevance distribution is incorporated into the attention logits as an adaptive positional bias, allowing each input and attention head to emphasize different local, periodic, or long-range dependencies. The overall adaptation framework is illustrated in Fig. 2.

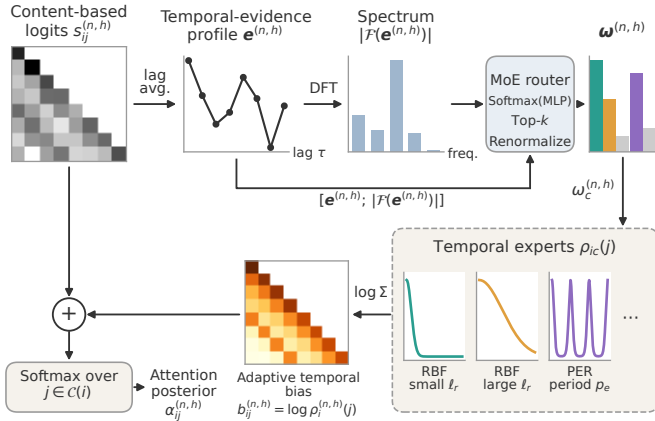


Fig. 2. **Architecture of the proposed Bayesian temporal inductive bias.** For the n th input and h th attention head, the attention posterior $\alpha_{ij}^{(n,h)}$ is obtained by adding an adaptive temporal bias $b_{ij}^{(n,h)}$ to the content-based logits $s_{ij}^{(n,h)}$ before softmax (left). Top: lag-averaged logits form the temporal-evidence profile $e^{(n,h)}$. The MoE router uses this profile and its Fourier magnitude spectrum $|\mathcal{F}(e^{(n,h)})|$ to produce top- k mixture weights, renormalized to sum to one. Bottom right: these weights combine normalized temporal experts $\rho_{ic}(j)$ constructed from RBF and PER kernels, yielding $\rho_i^{(n,h)}(j)$ and $b_{ij}^{(n,h)} = \log \rho_i^{(n,h)}(j)$. Different inputs and heads can thus emphasize local, long-range, or periodic dependencies.

A. Probabilistic Mixture of Temporal Kernels

For notational simplicity, we omit the input and attention-head indices (n, h) throughout this subsection. Each temporal expert defines a distribution over the key positions accessible to query i :

$$\rho_{ic}(j) = \frac{\exp[\psi^{(c)}(\tau_{ij}; \boldsymbol{\theta}^{(c)})]}{Z_i^{(c)}},$$

where $\tau_{ij} = |i - j|$ and

$$Z_i^{(c)} = \sum_{l \in \mathcal{C}(i)} \exp[\psi^{(c)}(\tau_{il}; \boldsymbol{\theta}^{(c)})].$$

Thus, $\sum_{j \in \mathcal{C}(i)} \rho_{ic}(j) = 1$. Different experts encode complementary temporal patterns, including locality, periodicity, and long-range dependence.

The MoE router combines these distributions using input- and head-specific weights:

$$\rho_i(j) = \sum_{c=1}^M \omega_c \rho_{ic}(j), \quad \omega_c \geq 0, \quad \sum_{c=1}^M \omega_c = 1.$$

The resulting ρ_i is a normalized temporal relevance distribution over accessible key positions.

The probabilistic mixture also provides a flexible mechanism for incorporating prior knowledge about the target dataset, such as daily or weekly seasonality. These kernels can be added to the candidate expert set without modifying the pretrained TSFM or the overall adaptation framework. The subsequent MoE router determines the contribution of each kernel for every input and attention head, allowing the model to exploit target-specific temporal priors without applying a single positional structure uniformly across all inputs.

B. MoE Router

To determine the temporal-kernel weights adaptively, we design an MoE router that extracts temporal evidence from the content-based attention scores, as illustrated in Fig. 2. For the h th attention head of the n th input, the original attention score between query i and key j is $s_{ij}^{(n,h)} = (\mathbf{q}_i^{(n,h)})^\top \mathbf{k}_j^{(n,h)} / \sqrt{d_k}$. We construct a one-dimensional temporal-evidence profile by averaging the two-dimensional attention scores associated with the same relative lag:

$$e_\tau^{(n,h)} = \frac{1}{|\mathcal{I}_\tau|} \sum_{(i,j) \in \mathcal{I}_\tau} s_{ij}^{(n,h)}, \quad \mathcal{I}_\tau = \{(i,j) : |i - j| = \tau\}.$$

This aggregation summarizes the content-based evidence for dependencies at different temporal lags. To capture both lag-domain and periodic information, the temporal-evidence profile and its Fourier spectrum are concatenated and fed into a multilayer perceptron:

$$\boldsymbol{\omega}^{(n,h)} = \text{Softmax}\left(\text{MLP}\left(\left[e^{(n,h)}; |\mathcal{F}(e^{(n,h)})|\right]\right)\right),$$

where $\mathcal{F}(\cdot)$ denotes the discrete Fourier transform and $\boldsymbol{\omega}^{(n,h)} = [\omega_1^{(n,h)}, \dots, \omega_M^{(n,h)}]$ contains the input- and head-specific mixture weights. We retain the k largest weights (top- k), set the remaining weights to zero, and renormalize the

retained weights to sum to one. Hereafter, $\omega^{(n,h)}$ denotes the resulting sparse mixture weights.

The router combines the predefined temporal kernels using these weights to construct the adaptive positional bias

$$b_{ij}^{(n,h)} = \log \left[\sum_{c=1}^M \omega_c^{(n,h)} \rho_{ic}(j) \right].$$

The resulting bias is added to the original attention scores before the softmax. Because the candidate kernels and their hyperparameters remain fixed during fine-tuning, the router learns to combine them without selecting a single kernel per input or repeatedly tuning kernel hyperparameters, yielding a lightweight, plug-and-play module for standard multi-head self-attention.

V. EXPERIMENTS

A. Experimental Setup

Dataset. We evaluate the proposed method on five widely used benchmarks covering different sampling resolutions, dimensionalities, and temporal patterns: ETTh1 and ETTh2 (7 variables, 17,420 hourly steps), ETTm1 and ETTm2 (7 variables, 69,680 steps at 15-minute resolution), and Weather (21 variables, 52,696 steps at 10-minute resolution).

Reference Models. We select a representative TSFM, namely Timer-XL [25], to evaluate the proposed adaptation method. In addition, four typical full-shot TSF models, PatchTST [40], iTransformer [41], DLinear [13], and Non-stationary Transformer [42], are selected as reference models.

Bayesian Temporal Inductive Bias Instantiation. We instantiate the temporal inductive bias using two families of time-aware kernels: *radial basis function (RBF)* and *periodic (PER)* kernels. Their log-kernel functions are defined as

$$\begin{aligned} \psi_r^{\text{RBF}}(\tau_{ij}; \theta_r^{\text{RBF}}) &= -\frac{\tau_{ij}^2}{2\ell_r^2}, \\ \psi_{r,e}^{\text{PER}}(\tau_{ij}; \theta_{r,e}^{\text{PER}}) &= -\frac{2\sin^2(\pi\tau_{ij}/p_e)}{(\ell'_{r,e})^2}, \end{aligned}$$

where $\tau_{ij} = |i - j|$ is the relative lag between query position i and key position j , measured in patches. Here, p_e denotes the selected dominant period for a dataset. All five PER experts share this period and differ only in their length scales; together with the ten RBF experts, they form $M = 15$ experts. The parameters ℓ_r and $\ell'_{r,e}$ control the length scales of the RBF and PER kernels, respectively. The corresponding kernel hyperparameters are $\theta_r^{\text{RBF}} = \{\ell_r\}$ and $\theta_{r,e}^{\text{PER}} = \{\ell'_{r,e}, p_e\}$.

To represent temporal dependencies ranging from short-term fluctuations to long-range interactions, we use multi-scale grids for both kernel families:

$$\begin{aligned} \ell_r &= 2^{r-9}L, \quad r = 1, \dots, 10, \\ \ell'_{r,e} &= 2^{r-9}p_e, \quad r = 1, \dots, 5, \end{aligned} \quad (3)$$

where L is the number of patches in the input sequence.

The dyadic spacing in Equation 3 covers temporal receptive fields at geometrically increasing scales. For an RBF kernel,

the positional penalty grows as $-\tau_{ij}^2/(2\ell_r^2)$; therefore, halving ℓ_r increases the magnitude of the penalty at a fixed lag by a factor of four. Each length scale consequently induces a different soft cutoff on the historical context, allowing the kernel bank to represent dependencies ranging from highly local to nearly global. For example, an input of 672 observations with a patch length of 96 contains $L = 7$ patches. At a 15-minute sampling resolution, each patch spans one day. With $\ell_r = L/16$, a four-day lag corresponds to $\tau_{ij} = 4$ and produces $-\tau_{ij}^2/(2\ell_r^2) \approx -41.8$. The corresponding kernel value, $\exp(-41.8)$, is effectively zero, strongly suppressing attention to lags of four days or longer. Larger length scales produce weaker decay and preserve progressively longer-range information. We cap the largest RBF length scale at $2L$, because $\ell_r \gg L$ produces an almost flat log-kernel over the input range and therefore provides little meaningful temporal inductive bias.

Implementation Details. We adopt the official repositories to implement all the aforementioned models. The input length is 672 unless otherwise stated. We evaluate LTSF performance using mean squared error (MSE) and mean absolute error (MAE), and report the number of learnable parameters to quantify model complexity. All experiments are implemented in PyTorch and executed on an NVIDIA A100 GPU with 80GB memory.

B. Main Results

Table I compares Timer-XL adapted with the proposed Bayesian temporal inductive bias against its zero-shot and full-shot variants. Averaged across five datasets and four forecasting horizons, the adapted Timer-XL achieves an MSE of 0.3194 and an MAE of 0.3544. Compared with zero-shot Timer-XL (0.3306/0.3654), it reduces MSE and MAE by 3.4% and 3.0%, respectively. It also improves over full-shot Timer-XL (0.3264/0.3616) by 2.1% in MSE and 2.0% in MAE. These improvements require only a modest 7.2% increase in parameter count, from 69M to 74M parameters.

In addition, Table I shows that Timer-XL equipped with the proposed Bayesian temporal inductive bias achieves the best MSE in 5 of 20 settings and the best MAE in 13 of 20 settings, outperforming the full-shot reference models in most MAE comparisons.

C. Case Study

To further examine the behavior of the proposed Bayesian temporal inductive bias, we adapt Timer-XL to ETTm1 using an input length of 2880 and a patch length of 96. Because ETTm1 is sampled every 15 minutes, the input covers 30 days and each patch represents one day. We select two input sequences and use zero-shot and adapted Timer-XL to forecast the next 96 time steps, i.e., one day. Fig. 3 presents the forecasts together with the selected temporal kernels and their router weights.

In both examples, adapted Timer-XL more closely follows the ground truth, reducing MSE from 0.019 to 0.012 and from 0.031 to 0.019 for the left and right sequences, respectively

TABLE I
MULTIVARIATE FORECASTING PERFORMANCE ON FIVE BENCHMARK DATASETS AND FOUR PREDICTION HORIZONS. WE COMPARE ZERO-SHOT, FULL-SHOT, AND ADAPTED TIMER-XL (WITH THE PROPOSED BAYESIAN TEMPORAL INDUCTIVE BIAS) WITH REPRESENTATIVE FULL-SHOT FORECASTING MODELS. LOWER MSE AND MAE INDICATE BETTER PERFORMANCE; THE BEST AND SECOND-BEST RESULTS IN EACH ROW ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY.

Model	Timer-XL (2025) [25]						iTransformer [41]	PatchTST [40]	DLinear [13]	Non-stationary [42]				
Training	Zero-shot		Full-shot		Adaptation		Full-shot		Full-shot		Full-shot		Full-shot	
Complexity	69M		69M		74M		–		–		–		–	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	0.369 0.391	0.364 0.397	0.343 0.379	0.387	0.418	0.373	0.403	0.369	0.400	0.452	0.478		
	192	0.405 0.413	0.405 0.424	0.376 0.402	0.416	0.437	0.405	0.425	0.405	0.422	0.484	0.510		
	336	0.418 0.423	0.427 0.439	0.391 0.413	0.434	0.450	0.423	0.440	0.435	0.445	0.511	0.522		
	720	0.423 0.441	0.439 0.459	0.406 0.433	0.447	0.473	0.445	0.471	0.493	0.508	0.571	0.543		
	Avg.	0.404 0.417	0.409 0.430	0.379 0.407	0.421	0.445	0.412	0.435	0.426	0.444	0.505	0.513		
ETTm2	96	0.283 0.342	0.277 0.343	0.289 0.345	0.304	0.362	0.289	0.347	0.305	0.371	0.348	0.403		
	192	0.340 0.379	0.348 0.391	0.350 0.389	0.372	0.407	0.360	0.393	0.412	0.439	0.408	0.448		
	336	0.366 0.400	0.375 0.418	0.371 0.411	0.418	0.440	0.389	0.420	0.527	0.508	0.424	0.457		
	720	0.397 0.431	0.409 0.458	0.431 0.459	0.463	0.476	0.398 0.440	0.830	0.653	0.448	0.476			
	Avg.	0.347 0.388	0.352 0.402	0.360 0.401	0.389	0.421	0.359 0.400	0.518	0.493	0.407	0.446			
ETTh1	96	0.317 0.356	0.290 0.341	0.284 0.337	0.311	0.365	0.285 0.346	0.307	0.350	0.414	0.414			
	192	0.358 0.381	0.337 0.369	0.332 0.367	0.353	0.390	0.329 0.372	0.337 0.368	0.524	0.482				
	336	0.386 0.401	0.374 0.392	0.367 0.390	0.387	0.411	0.363 0.394	0.366 0.387	0.541	0.497				
	720	0.430 0.431	0.437 0.428	0.430 0.426	0.452	0.445	0.421 0.426	0.419 0.419	0.578	0.509				
	Avg.	0.373 0.392	0.359 0.382	0.353 0.380	0.376	0.403	0.349 0.385	0.357 0.381	0.514	0.475				
ETTm2	96	0.189 0.277	0.175 0.257	0.173 0.252	0.183	0.272	0.172 0.259	0.167 0.263	0.237	0.306				
	192	0.241 0.315	0.242 0.301	0.235 0.294	0.250	0.315	0.233 0.299	0.230 0.311	0.330	0.387				
	336	0.286 0.348	0.293 0.337	0.291 0.331	0.311	0.356	0.280 0.331	0.298 0.361	0.404	0.424				
	720	0.375 0.402	0.376 0.390	0.382 0.396	0.417	0.419	0.357 0.382	0.432 0.446	0.525	0.486				
	Avg.	0.273 0.336	0.272 0.321	0.270 0.318	0.290	0.340	0.261 0.318	0.282 0.345	0.374	0.401				
Weather	96	0.171 0.225	0.157 0.205	0.153 0.197	0.174	0.225	0.149 0.202	0.169	0.229	0.185	0.241			
	192	0.221 0.271	0.206 0.250	0.200 0.243	0.227	0.268	0.194 0.245	0.211	0.268	0.286	0.325			
	336	0.274 0.311	0.259 0.291	0.255 0.285	0.290	0.309	0.244 0.285	0.258	0.306	0.323	0.347			
	720	0.356 0.370	0.337 0.344	0.332 0.338	0.374	0.360	0.317 0.338	0.320	0.362	0.436	0.401			
	Avg.	0.256 0.294	0.240 0.273	0.235 0.266	0.266	0.291	0.226 0.268	0.239	0.291	0.308	0.329			
1st Count	3	4	1	0	5	13	0	0	8	4	3	2	0	0

(36.8% and 38.7% reductions). More importantly, the router selects different temporal structures for different inputs. The left series undergoes a sharp decrease during the two days immediately preceding the forecasting origin, making earlier observations less representative of the current regime; consistently, the router assigns its largest weights to short-range locality kernels. The right series exhibits no comparable recent regime change, and the router assigns greater weights to long-range locality and periodic kernels, which draw on a broader historical context. This case study illustrates how adaptive PE can adjust the temporal-evidence likelihood to input-specific non-stationarity when adapting a TSFM.

D. Hyperparameter Analysis

The Bayesian temporal inductive bias has two main hyperparameters: the number of kernels assigned nonzero routing

weights, denoted by $\text{top-}k$, and the hidden dimension of the MLP router, denoted by d_{router} . We examine their effects on the ETTh1 validation MSE at forecasting horizon $H = 192$ through grid search. Specifically, $\text{top-}k$ is selected from $\{3, 5, 7, 9\}$, while d_{router} is varied from 8 to 488 with an interval of 32.

Fig. 4(a) shows that $\text{top-}k = 5$ achieves the lowest validation MSE (0.92665). Using three kernels may limit temporal-pattern diversity, whereas seven or nine increase the MSE by approximately 0.6%. Fig. 4(b) shows that the validation MSE, averaged over $\text{top-}k$, is minimized at $d_{\text{router}} = 40$ (approximately 0.9298). Larger routers yield no consistent improvement, with relative MSE increases of up to approximately 0.3%, suggesting that a moderate router capacity suffices.

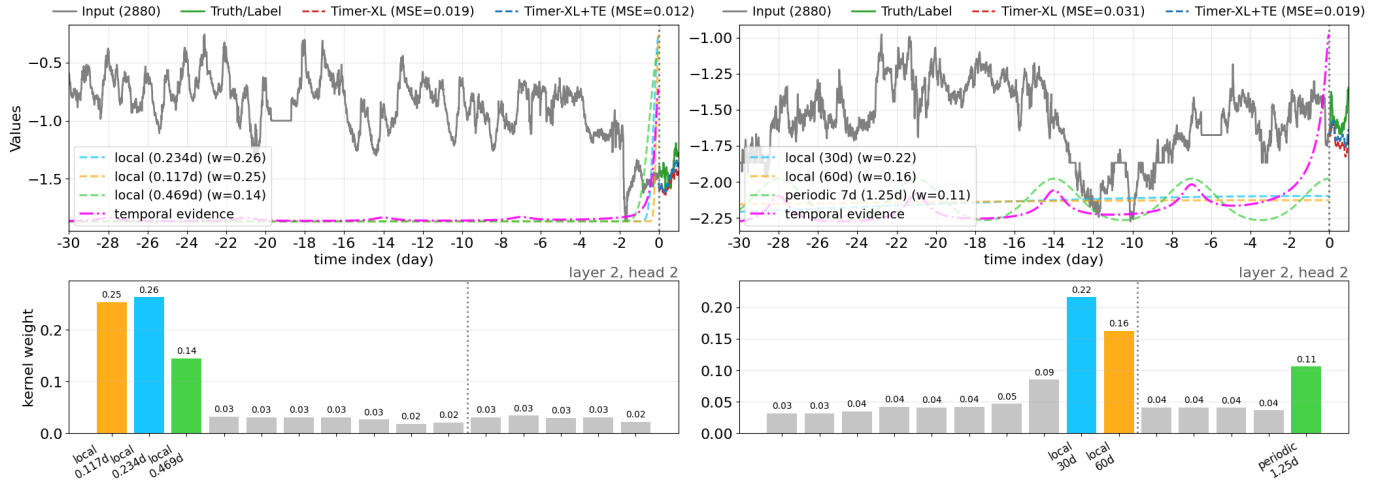


Fig. 3. **Case study of adaptive temporal-pattern selection on ETTh1.** Two 30-day inputs are used to forecast the following day. Top: predictions of zero-shot and adapted Timer-XL, with the adapted model’s temporal kernels; bottom: router weights for layer 2, head 2. The sharp regime change just before the forecasting origin of the left input makes earlier observations less representative, consistent with the router’s preference for short-range locality kernels. For the more stable right input, the router favors long-range locality and periodic kernels, drawing on a broader historical context.

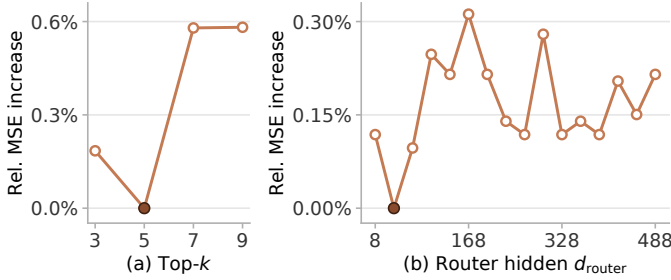


Fig. 4. Hyperparameter sensitivity on ETTh1 ($H = 192$). Relative validation MSE increases are measured against each panel’s best setting (filled marker): (a) top- k at $d_{\text{router}} = 40$; (b) d_{router} , averaged over top- k .

TABLE II
ABLATION STUDY OF BAYESIAN TEMPORAL INDUCTIVE BIAS ON ETTh1.

	Model	Zero-shot Timer-XL		Adapted Timer-XL			
		RoPE		MoE (Local)		MoE (Local+Periodic)	
	Metric	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	0.369	0.391	0.356	0.384	0.343	0.379
	192	0.405	0.413	0.389	0.406	0.376	0.402
	336	0.418	0.423	0.405	0.417	0.391	0.413
	720	0.423	0.441	0.413	0.438	0.406	0.433
	Avg.	0.404	0.417	0.391	0.411	0.379	0.407
	Gain (%)	–	–	3.22	1.38	6.13	2.46

E. Ablation Study

To assess the contributions of different temporal kernels, we conduct an ablation study on ETTh1. We compare two variants of the proposed Bayesian temporal inductive bias: an MoE router using (i) only local RBF kernels and (ii) both local RBF and periodic kernels. Zero-shot Timer-XL is included as a reference. Table II summarizes the results.

The local-only variant consistently outperforms zero-shot Timer-XL across all forecasting horizons, reducing the average MSE and MAE by 3.22% and 1.38%, respectively. This result indicates that introducing temporal locality biases improves the model’s ability to capture short-range dependencies [43]. Incorporating periodic kernels yields further improvements: the complete model achieves the best performance at every forecasting horizon, with average MSE and MAE reductions of 6.13% and 2.46%, respectively, relative to zero-shot Timer-XL. These results suggest that local and periodic kernels play complementary roles in capturing heterogeneous temporal dependencies.

VI. CONCLUSIONS

We presented a Bayesian interpretation of attention that connects content-based similarity with positional temporal weighting. Motivated by the limitations of static lag-based biases, we proposed an adaptive mixture of temporal kernels, routed using lag-domain and Fourier features. On five benchmarks, adapted Timer-XL reduces average MSE and MAE by 3.4% and 3.0% over its zero-shot variant and by 2.1% and 2.0% over its full-shot variant, respectively, with a 7.2% increase in parameter count.

This study has several limitations that motivate future work. The evaluation is limited to Timer-XL and five datasets, four from the ETT family. The fixed kernel bank may also inadequately represent irregular, abrupt, or evolving temporal structures. Future directions include evaluation across encoder-only, decoder-only, and encoder-decoder TSFMs, including Moirai, Chronos, Time-MoE, and Sundial; broader benchmarks spanning energy, traffic, finance, and climate; and learnable or automatically constructed temporal kernels to improve the method’s flexibility under complex distribution shifts.

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